

HANDYMAN HELPER

Final Project Report

University Of Regina

ENSE 281 - Software Project Management

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Team Members

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Github Repository: <https://github.com/TGoncin/HandymanHelper.git>

1. Summary

Handyman Helper is a digital marketplace designed to restore transparency and fairness to home-service trades. The platform directly connects local contractors with clients, removing pay-to-play advertising barriers and replacing them with a verified credential system and a competitive direct-bidding model.

Over the course of this project, our team of four designed, prototyped, and delivered a working Minimum Viable Product (MVP) built on a MERN-lite architecture ([Node.js](#) / Express, MongoDB, EJS templating). The MVP supports two distinct user roles, Client and Contractor, and enforces core business rules around verified bidding, job-lifecycle management, and mobile-first usability.

This report documents our exploration process, key decisions, stakeholder feedback, technical deliverables, and the lessons we've learned along the way.

2. Project Goals & Objectives

Our team set out to address a genuine pain point in the home-services market. We defined the following goals at the outset of the project:

- Launch a functional MVP: Connect verified contractors with local clients through a working web application.
- Implement credential verification: Allow contractors to upload certifications and/or link an instagram portfolio before becoming searchable.
- Build transparent review features: Two-way ratings unlocked only after a deal is struck, preventing bot manipulation.
- Ensure scalability: Design the data models and architecture to support future expansion to additional cities and provinces.
- Measure success: User growth, verified provider adoption, client satisfaction scores, and successful job completion metrics.

Were the Goals Achieved?

For the most part, yes, the core MVP goals were met. The bidding system, role-based access control, verification gate, and job-lifecycle management (OPEN to CLOSED) all function as specified.

The rating system and full category filtering remain as next-step enhancements, having been descoped from the MVP to keep the initial release focused.

3. Team Exploration & Process

Our process followed an iterative, user-centered approach informed by agile principles. The high-level phases were:

3.1 Discovery & Scoping

We began by researching existing platforms such as Kijiji, Facebook Marketplace, TaskRabbit etc., to understand what they did well and where they failed. The recurring theme was that contractors were forced to pay for visibility, not quality. This insight became the core value proposition of Handyman Helper.

During scoping, we produced the following planning artifacts stored in our GitHub Project Archive:

- Business Case - where we justified the market opportunity
- Project Charter - defined the scope, constraints, and stakeholder sign-off
- Project Requirements Document - defined functional and non-functional requirements
- Project Scope Statement - showed the explicit in-scope and out-of-scope definitions
- Stakeholder Register & Engagement Plan - identifying clients, contractors and admin personas

3.2 Low-Fidelity Prototyping

Before writing a single line of code, we produced paper and PDF wireframes for:

- Login & Verification flow
- Main Search / Marketplace page
- Post Job & Job Details screens

These prototypes allowed us to test navigation assumptions cheaply and gather early feedback.

3.3 MVP Definition & Development

The formal MVP Definition (stored in our GitHub archive) specifies the exact acceptance criteria for each feature. The development sprint delivered the following:

- Architectural migration from a flat .JSON file system to MongoDB + Mongoose for database storage.
- [Express.js](#) server with protected middleware.
- Role-based registration with an invite-code gate to limit access during testing.
- Marketplace with skip/limit pagination (25 ads per page) and case-insensitive Regex search.
- Multer-powered license photo upload for contractor verification.
- Master Switch: unverified contractors cannot place bids.
- Job OPEN to CLOSED lifecycle managed by the job-owning client.

3.4 Project Management

We managed work using a GitHub KanBan board (linked from our README) and held regular team sync meetings documented in our Group Reflections. A RACI Chart and Communication Management Plan governed our decision-making responsibilities throughout.

4. Stakeholder Feedback & Impact

Feedback collection was built directly into our product. A GitHub Discussion poll linked from the README, with a QR code on the landing page, invited users to share their experiences. We also gathered structured feedback through peer evaluations and group reflections at two sprint checkpoints.

4.1 Feedback Themes

- Verification trust: Stakeholders (Dr. Maciag) consistently emphasized that the verification feature had to be handled with care, and is important. It made the platform feel credible.
- Mobile experience: Members of the group, and some external testers noted the interface was easy to navigate on a phone, which validates the mobile-first decision.
- Invite code friction: At first, some members found the registration invite code confusing, which was a very fair reaction. We acknowledged this as an intentional testing gate, not a permanent design choice.
- Bidding clarity: Some users wanted clearer feedback when a bid was submitted successfully. This informed a UX refinement in the final sprint.

4.2 How Feedback Changed the Project

The verification-first emphasis confirmed that the Master Switch (blocking unverified contractor bids) was the right call to prioritize. Feedback on search usability led to the addition of a “Clear Search” button (noted in the MVP next steps). The general enthusiasm for the direct-bidding concept reinforced our decision to keep that feature central rather than descope it. We agreed, however, to descope the ratings system from the MVP. This trade-off was well-received in the final demo, it seems.

5. Technical Deliverables

5.1 Architecture

The final MVP runs on a MERN-lite stack:

Component	Technology	Purpose
Runtime	Node.js/Express.js	Handling server-side logic and routing.
Database	MongoDB	Replaced flat .json files, ensuring data persistence.
ODM	Mongoose	Defining User and Job schemas.
Templating	EJS	Rendering dynamic HTML based on server-side state.
File Handling	Multer	Managing contractor license uploads.

5.2 Data Models

User Model (models/[User.js](#))

Stores authentication credentials, role (CLIENT | CONTRACTOR), verification state, ID photo path, Instagram handle, license number, and a ratings array. The ratings array is designed to gate review visibility until a job is closed.

Job Model (models/[Job.js](#))

Stores job title, description, budget, category, owning client username, status (OPEN | CLOSED), and a bids array. The timestamps option provides automatic createdAt and updatedAt fields for audit purposes.

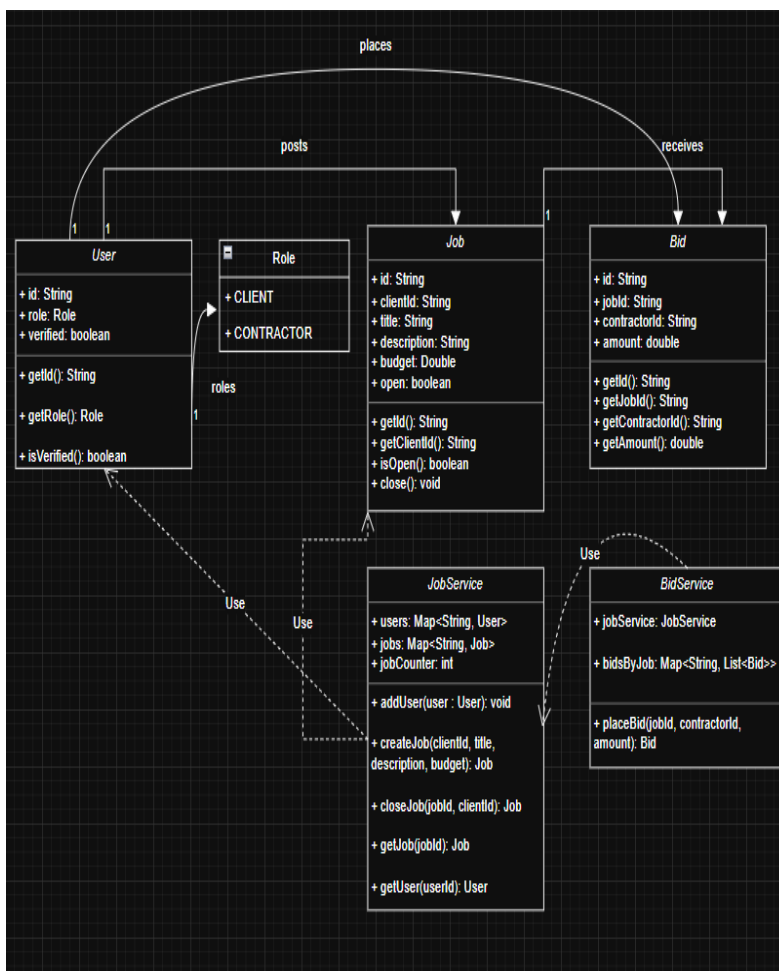
5.3 Key Route Logic

- POST/register: Validates invite code, checks for duplicate usernames, creates user with role assignment.
- POST/marketplace: Requires login (isLoggedIn middleware), applies Regex search and skip/limit pagination, preserves currentUser and searchQuery state via a hidden pageForm.
- POST/bid: jobId: Verifies user is a contractor, job is open, and contractor isVerified before appending bid.
- POST/close/: jobId: Verifies user is a client and owns the job before transitioning status to closed.

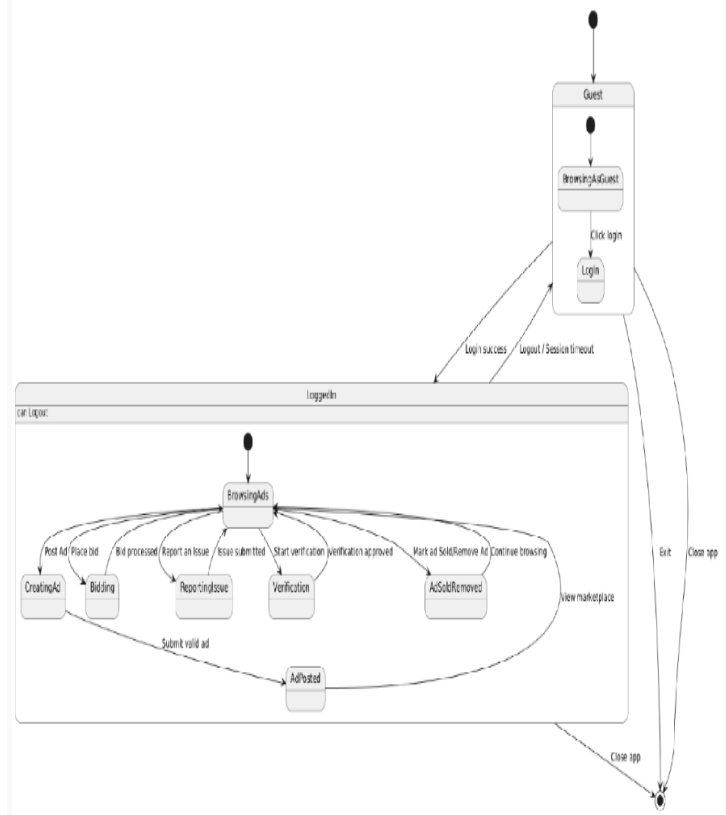
- POST/upload-license: Uses Multer to receive a photo file, saves to /public/uploads, and updates idPhotoPath on the User document.

5.4 UML & System Diagrams

A Class Diagram and State Diagram are included in the GitHub archive under /Project Archive & Histroy/UML & Diagram/. The State Diagram in particular captures the Open to Closed job lifecycle and the Unverified to Verified contractor transition.



Handyman Helper— State Diagram



6. Key User Interactions (MVP)

The delivered MVP supports the following end-to-end interactions for each persona:

6.1 Client Journey

- Register with invite code and select the client role
- Browse the marketplace
- Post a new job with title, description, and budget
- Review bids submitted by verified contractors
- Close the job once a contractor is selected and work is complete

6.2 Contractor Journey

- Register with invite code and select the contractor role
- Upload license photo/invite code
- Once verified, browse the marketplace
- Submit competitive bids on open jobs

7. Team Reflection

The following reflection was developed collaboratively as a team and represents our honest assessment of the project experience.

7.1 How did we feel about this Project (as a whole)?

We didn't like how the most important part of the project came at the very end. We understand there's no real way around it, but having 9 labs, a project, and a final exam all at once while managing other classes felt pretty brutal. It takes away from the enjoyment that should come with building something meaningful. The project itself was cool, and having others come to watch it in action made it even more valuable. However, we learned that teams need to meet in person to keep things moving. When people aren't as invested, progress slows. Nonetheless, everything came together in the end and everyone pulled their fair share.

7.2 What is the team most proud?

We are proud of the achieving the MVP in a clear and manageable manner. Our team overcame a lot. Without getting into personal details, there were many variables that seemed to work against us. However, by the end things came together and we successfully completed our MVP.

7.3 What did we learn about ourselves?

Individually and as a team, we learned that accountability and involvement are KEY. The standards we set for ourselves become the standards for the entire team. It creates a sort of domino effect, as one person improves, others follow. Working efficiently together requires being fast, flexible, and ready to pivot. If one person falls behind, the whole team feels it. To achieve a quality outcome, we need to act as a team and support one another.

7.4 How will we use (or not use) what we learned in this activity going forward?

We will definitely carry forward the web development experience from Labs 4-9, along with the documentation work we completed, as it was extremely valuable. We will also continue organizing projects into their own GitHub repositories, since that made everything more structured and manageable.

8. Next Steps & Future Work

The following enhancements are documented in the MVP Development Report as the immediate roadmap beyond the MVPs:

- **Admin Panel:** A UI to flip the isVerified flag for new contractors, replacing the current database-direct approach.
- **Clear Search Button:** A UI control in the sticky menu to reset the search query and return to the full marketplace listing.
- **Ratings UI:** Surface the ratings array with a front-end form unlocked post-job-close.
- **Geographic Expansion:** Add a city/province field to Job and User models to enable location-based filtering.

9. Screenshots of Application

The following screenshots were captured from the live MVP.

They illustrate the four primary views delivered in the final build.

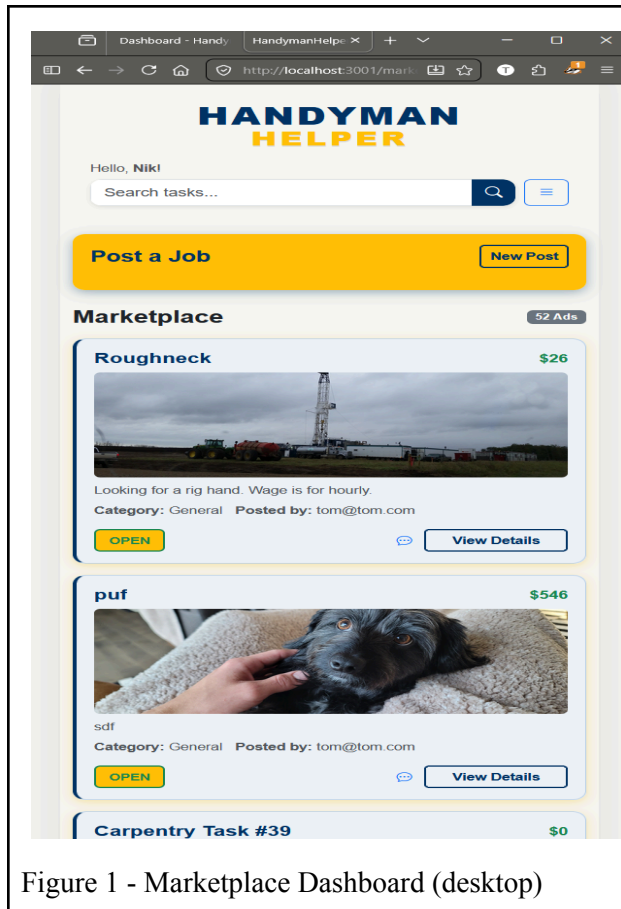


Figure 1 - Marketplace Dashboard (desktop)

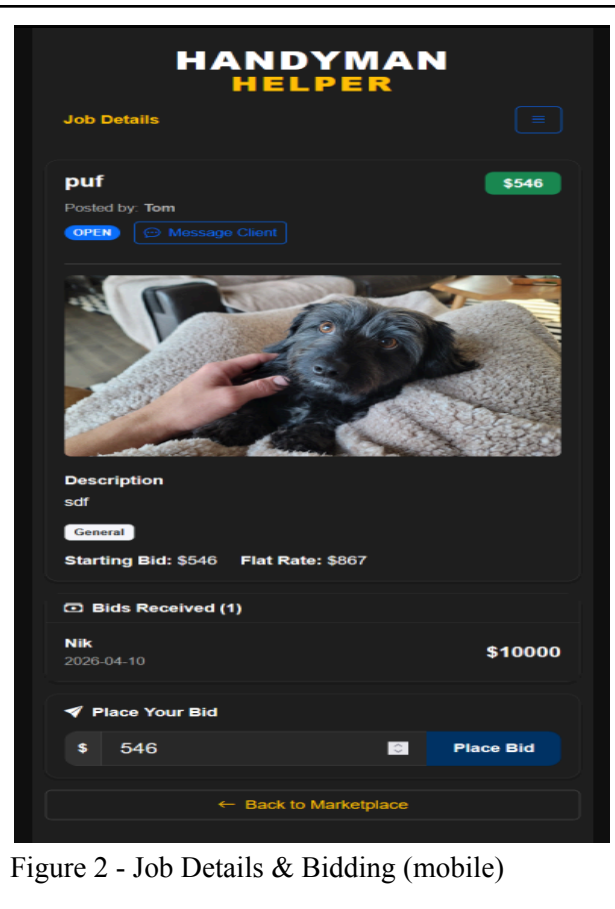


Figure 2 - Job Details & Bidding (mobile)

HANDYMAN HELPER

Choose one of two options to get verified:

Submit your credentials to unlock bidding on jobs.

First Name *

This is required for verification.

Option 1: Upload License Photo

No file selected.

Take a photo or upload your trade/craft/skill certificate or license

OR

Option 2: Invite Code

If you've received the invite code, please enter it below:

Figure 3 - Contractor Verification Page

HANDYMAN HELPER

Welcome, Nik

This is your personal dashboard.

My Messages

Conversation with tom@tom.com

Is it only the dog?

Does it come with that blanket.. ? It looks like a comfy blanket.. and he looks comfy in it.

2026-04-10, 1:26:23 p.m.

Re: Is it only the dog?

HELLLLLLLOOOOOOOOOOOOOOO

2026-04-10, 1:27:06 p.m.

Re: Is it only the dog?

Just because of your attitude.. you blacklisted.. & IT DID COME WITH THE BLANKET... You snooze you lose!

2026-04-10, 1:27:53 p.m.

Re: Is it only the dog?

YOUR GOING TO REGRET THIS

2026-04-10, 1:28:26 p.m.

Re: Is it only the dog?

Currently Open Bids

purf Client: tom@tom.com OPEN	\$546 My Budget: \$50000
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Figure 4 - Messaging & Dashboard